TOPIC 02 · STRUCTURE OF MATERIALS AND CRYSTALLINE DEFECTS

Structure of Materials and Crystalline Defects

Crystal lattices, the BCC, FCC and HCP structures, how to name positions, directions and planes, atomic densities, and the defects of real crystals.

Source: Universidad Carlos III de Madrid · Session 3 T2 Structure of Materials I_2025.pdf and Session 4 T2 Structure of Materials II_2025.pdf

Central idea: the performance of a component depends on the properties of its material, and those properties depend on its **structure**, the way its atoms are arranged. This topic describes that arrangement (crystal lattices, the main metallic structures, how to name positions, directions and planes) and the defects that real crystals always contain.

Part I covers the perfect crystal: crystal systems, atomic packing, the BCC, FCC and HCP structures, interstitial sites, and how to describe positions, directions and planes. **Part II** covers densities, defects (point, linear and planar), solid solutions, and polymorphism.

Part I: Structure of materials

§ 1 Why the structure matters [Session 3 Slide 2]

The choice of a material for a part, such as an aircraft wing, depends on its properties: a ceramic like diamond would be too brittle, copper too weak, a rubber too flexible. Those properties are not random: they come from the **structure**, the arrangement of the components of the material.

$$\text{performance of a component} = f(\text{properties}), \qquad \text{properties} = f(\text{structure})$$

§ 2 Crystalline and amorphous solids [Session 3 Slides 3-4]

- **Crystalline solids** have a **periodic, long-range order**: the same arrangement of atoms repeats in all three directions. Most materials are crystalline: metals and alloys, and most ceramics.
- **Amorphous (non-crystalline) solids** have no periodic order and no symmetry over long distances. Glasses and some polymers are amorphous. A solid ends up amorphous only when, for some reason, it cannot form a crystal.

Why atoms order themselves: atoms tend to sit in the positions of **minimum energy**. A dense, ordered packing has lower energy than a loose, random one, and the result is a uniform, repeating arrangement called the **crystal lattice**.

§ 3 Lattice and unit cell [Session 3 Slides 5-7]

- The **crystal lattice** is an infinite three-dimensional arrangement of points.
- The **unit cell** is the smallest piece of the lattice that still contains all the symmetry elements of the crystal. It is a parallelepiped that, repeated over and over, rebuilds the whole crystal: the "master model" of the structure.

The idea goes back to Haüy (1781): however small he broke a calcite crystal, the pieces were always rhombohedral and fitted together with no gaps. So the crystal had to be made of identical small units packed perfectly. That raises the question of how many different shapes can fill space completely with no voids.

§ 4 Lattice parameters and the seven crystal systems [Session 3 Slide 8]

A unit cell is defined by three **lattice vectors a, b and c** that start at a common origin. It needs **six lattice parameters**:

- the three **edge lengths** a, b, c;
- the three **angles** between them, α, β, γ.

Classifying cells by these parameters gives **seven crystal systems**:

System	Edges	Angles
Cubic	$a = b = c$	$\alpha = \beta = \gamma = 90^\circ$
Hexagonal	$a = b \neq c$	$\alpha = \beta = 90^\circ,\ \gamma = 120^\circ$
Tetragonal	$a = b \neq c$	$\alpha = \beta = \gamma = 90^\circ$
Rhombohedral	$a = b = c$	$\alpha = \beta = \gamma \neq 90^\circ$
Orthorhombic	$a \neq b \neq c$	$\alpha = \beta = \gamma = 90^\circ$
Monoclinic	$a \neq b \neq c$	$\alpha = \gamma = 90^\circ \neq \beta$
Triclinic	$a \neq b \neq c$	$\alpha \neq \beta \neq \gamma \neq 90^\circ$

§ 5 Bravais lattices [Session 3 Slides 9-14]

The crystal system fixes the **shape** of the cell, but not **where the atoms sit** in it. Bravais (1848) showed that atoms can be placed in these seven systems in only **14 different ways**: the **Bravais lattices**.

There are four ways of placing atoms in a cell:

- **Simple (P):** one atom per lattice point, at the corners.
- **Body-centred (I):** an extra atom at the centre of the cell.
- **Face-centred (F):** extra atoms at the centre of every face.
- **Base-centred (C):** extra atoms at the centres of the two bases.

Combining the 7 systems with these 4 types gives the 14 Bravais lattices:

System	Bravais lattices
Cubic	simple, body-centred, face-centred

System	Bravais lattices
Tetragonal	simple, body-centred
Orthorhombic	simple, base-centred, body-centred, face-centred
Rhombohedral	simple
Hexagonal	simple
Monoclinic	simple, base-centred
Triclinic	simple

§ 6 Atomic packing [Session 3 Slides 16-22]

Imagine the atoms as hard spheres arranged in flat layers:

- In a **non-compact** plane each sphere sits in a square grid and touches only four neighbours; there are large gaps.
- In a **close-packed** plane each row is shifted so that every sphere sits in the hollow between two spheres of the next row; each sphere touches six neighbours and the gaps are as small as possible.

Stacking **non-compact** layers can still give a compact structure: BCC is built this way, with each layer sitting over the gaps of the one below.

Stacking **close-packed** layers can be done in two ways, because the third layer has two choices of hollows to sit in:

- **A B A B …** : the third layer sits exactly over the first. This is the **hexagonal close-packed (HCP)** structure.
- **A B C A B C …** : the third layer sits over neither of the first two; only the fourth repeats the first. This is the **face-centred cubic (FCC)** structure.

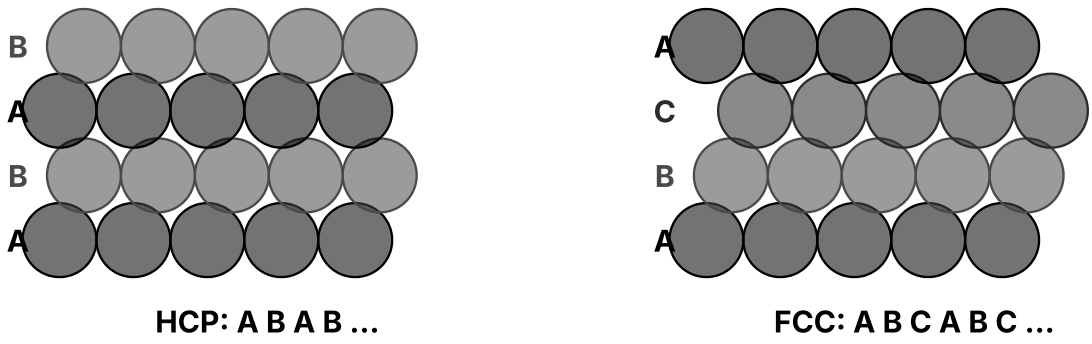


Figure 1. Stacking sequences of close-packed planes.

§ 7 Main metallic structures [Session 3 Slides 24–28]

About **90 % of metals** crystallize in one of three densely packed structures. They do so because:

- a pure metal is made of atoms of one element, so all atoms have the **same radius**;
- the metallic bond is **non-directional**, so atoms simply pack as closely as they can;
- smaller distances between atoms give **higher lattice energies**, so dense packing is the arrangement of minimum energy.

Structure	Examples
Body-centred cubic (BCC)	Fe, Cr, Mo, V, W, K
Face-centred cubic (FCC)	Al, Cu, Pb, Ni, Pt, Ag
Hexagonal close-packed (HCP)	Ti, Mg, Cd, Co, Zn

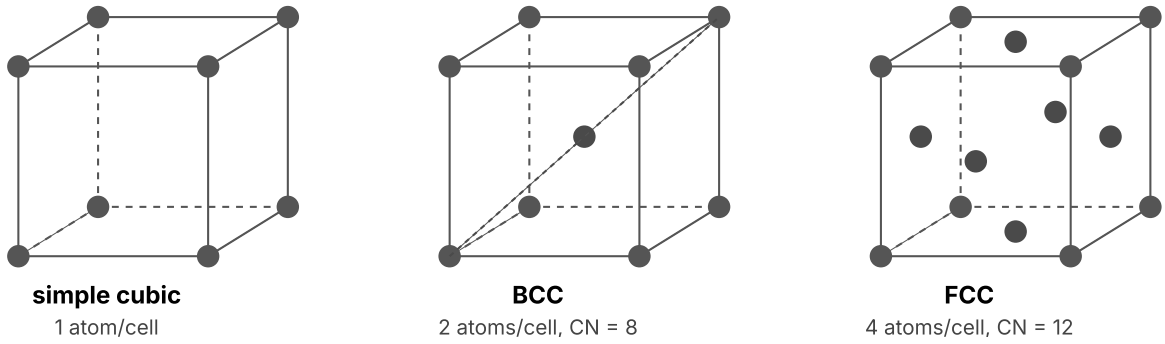


Figure 2. Simple cubic, BCC and FCC unit cells.

To compare them we use three numbers: how many atoms belong to one cell, how many neighbours each atom touches (the **coordination number**), and how much of the cell volume the atoms fill:

$$\text{Atomic Packing Factor (APF)} = \frac{\text{volume of the atoms in a unit cell}}{\text{total volume of the unit cell}}$$

Counting atoms per cell. An atom at a corner is shared by the 8 cells that meet there, so only ⅛ of it belongs to each cell. An atom on a face is shared by 2 cells (½ each). An atom inside the cell belongs to it completely.

7.1 Body-centred cubic (BCC) [Session 3 Slide 26]

- **Atoms per cell:** 8 corners × ⅛ + 1 centre = **2 atoms**.
- **Coordination number: 8** (the central atom touches the 8 corners).
- **Where atoms touch:** along the **body diagonal** of the cube. The face diagonal is d, with $d^2 = a^2 + a^2 = 2a^2$; the body diagonal D satisfies $D^2 = a^2 + d^2 = 3a^2$, so $D = \sqrt{3}\,a$. Along it lie a corner atom, the central atom and the opposite corner atom: $D = 4R$. Therefore

$$4R = \sqrt{3}\,a \quad \Rightarrow \quad a = \frac{4R}{\sqrt{3}}$$

- **Packing factor:**

$$APF = \frac{2 \cdot \frac{4}{3}\pi R^3}{\left(\frac{4R}{\sqrt{3}}\right)^3} = \frac{\frac{8}{3}\pi R^3}{\frac{64R^3}{3\sqrt{3}}} = \frac{\sqrt{3}\,\pi}{8} = 0.68$$

68 % of the cell is occupied by atoms.

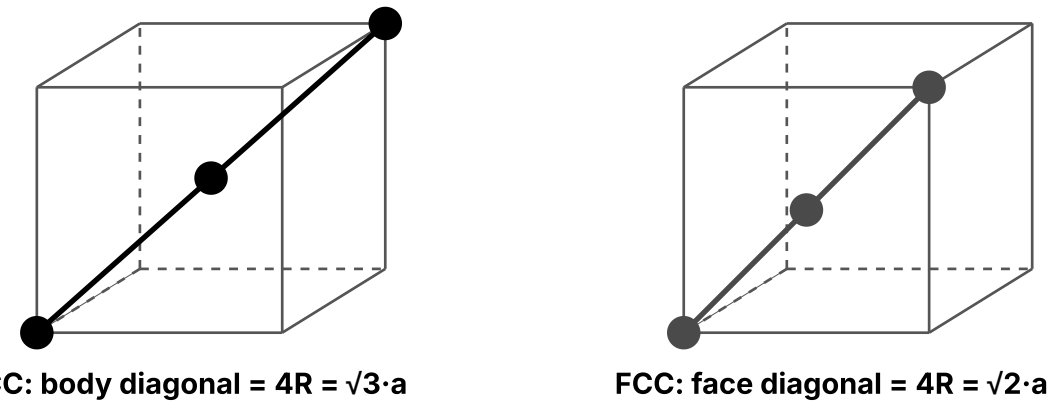


Figure 3. Where the atoms touch in BCC and FCC.

7.2 Face-centred cubic (FCC) [Session 3 Slide 27]

- **Atoms per cell:** 8 corners × ⅛ + 6 faces × ½ = **4 atoms**.
- **Coordination number: 12.**
- **Where atoms touch:** along the **face diagonal**: a corner atom, the face-centre atom and the opposite corner atom, so $d = 4R$. With $d^2 = a^2 + a^2$, $d = \sqrt{2}\,a$:

$$4R = \sqrt{2}\,a \quad \Rightarrow \quad a = 2\sqrt{2}\,R$$

- **Packing factor:**

$$APF = \frac{4 \cdot \frac{4}{3}\pi R^3}{\left(2\sqrt{2}\,R\right)^3} = \frac{\frac{16}{3}\pi R^3}{16\sqrt{2}\,R^3} = \frac{\pi}{3\sqrt{2}} = 0.74$$

74 % of the cell is occupied: FCC is a close-packed structure.

7.3 Hexagonal close-packed (HCP) [Session 3 Slide 28]

- **Atoms per cell:** 12 corners × ⅙ + 2 base centres × ½ + 3 inside = **6 atoms** (each corner of the hexagonal prism is shared by 6 cells).
- **Coordination number: 12.**
- **Geometry:** atoms touch along the edge of the hexagon, so $2R = a$. The height of the cell is $c = \frac{2\sqrt{2}}{\sqrt{3}}\,a$.
- **Volume:** the base is a hexagon made of 6 equilateral triangles of side a: $B = 3 \cdot \frac{\sqrt{3}}{2}a^2$. Then

$$V_{cell} = B \cdot c = \frac{3\sqrt{3}}{2}a^2 \cdot \frac{2\sqrt{2}}{\sqrt{3}}a = 3\sqrt{2}\,a^3 = 3\sqrt{2}\,(2R)^3 = 24\sqrt{2}\,R^3$$

- **Packing factor:**

$$APF = \frac{6 \cdot \frac{4}{3}\pi R^3}{24\sqrt{2}\,R^3} = \frac{8\pi}{24\sqrt{2}} = \frac{\pi}{3\sqrt{2}} = 0.74$$

Like FCC, 74 % of the cell is occupied. That is expected: FCC and HCP are the two ways of stacking the same close-packed layers.

§ 8 Interstitial sites [Session 3 Slides 30-35]

Even in the densest packing, 26 % of the volume is empty. The gaps between the regular atom positions are **interstitial sites**: places that other (usually smaller) atoms or ions can occupy. There are two kinds:

- **tetrahedral site:** surrounded by **4** lattice atoms (at the corners of a tetrahedron);
- **octahedral site:** surrounded by **6** lattice atoms (at the corners of an octahedron).

In compact structures, with n atoms per cell:

$$\text{tetrahedral sites} = 2n, \qquad \text{octahedral sites} = n$$

FCC ($n = 4$):

- tetrahedral: 8 sites completely inside the cell, 8 × 1 = **8**;
- octahedral: 1 at the centre of the cell, plus 12 at the middle of the edges, each shared by 4 cells: 1 + 12 × ¼ = **4**.

BCC ($n = 2$):

- tetrahedral: 4 on each of the 6 faces, each shared by 2 cells: 6 × 4 × ½ = **12**;
- octahedral: one at the centre of each face (6 × ½ = 3) plus one at the middle of each edge (12 × ¼ = 3): **6**.

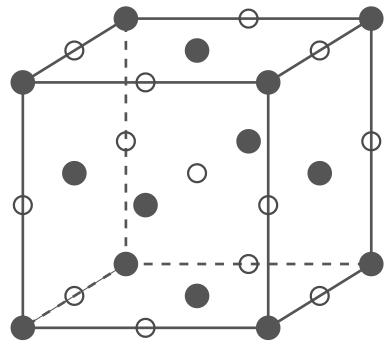
HCP ($n = 6$): **12** tetrahedral and **6** octahedral sites.

Structure	Atoms per cell	Tetrahedral sites	Octahedral sites
FCC	4	8	4
HCP	6	12	6
BCC	2	12	6

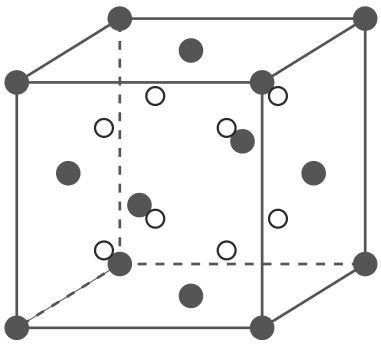
BCC is not close-packed, so the rule $2n$ / n does not apply to it.

NOTE ON THE OFFICIAL MATERIAL

For BCC, the 3 octahedral sites counted as "6 × ½" are labelled "center"; they are the centres of the 6 faces (each shared by 2 cells), not the centre of the cube.



FCC octahedral sites:
centre (1) + 12 edges × ¼ = 4



FCC tetrahedral sites:
8 inside the cell = 8

Figure 4. Octahedral and tetrahedral interstitial sites in FCC.

§ 9 Atomic positions [Session 3 Slide 37]

In an **orthogonal** lattice (all angles 90°), the three lattice vectors define every point of the cell without ambiguity, so they can be used to give the position of each atom. In a cubic cell ($a = b = c$) a position is written as three coordinates, each a fraction of the edge between **0 and 1**:

$$\text{position} = (x, y, z), \qquad 0 \le x, y, z \le 1$$

The origin is $(0, 0, 0)$; the corner along y is $(0, 1, 0)$; the top corner above the origin is $(0, 0, 1)$; the corner opposite the origin is $(1, 1, 1)$; the centre of the cube is $(\tfrac{1}{2}, \tfrac{1}{2}, \tfrac{1}{2})$.

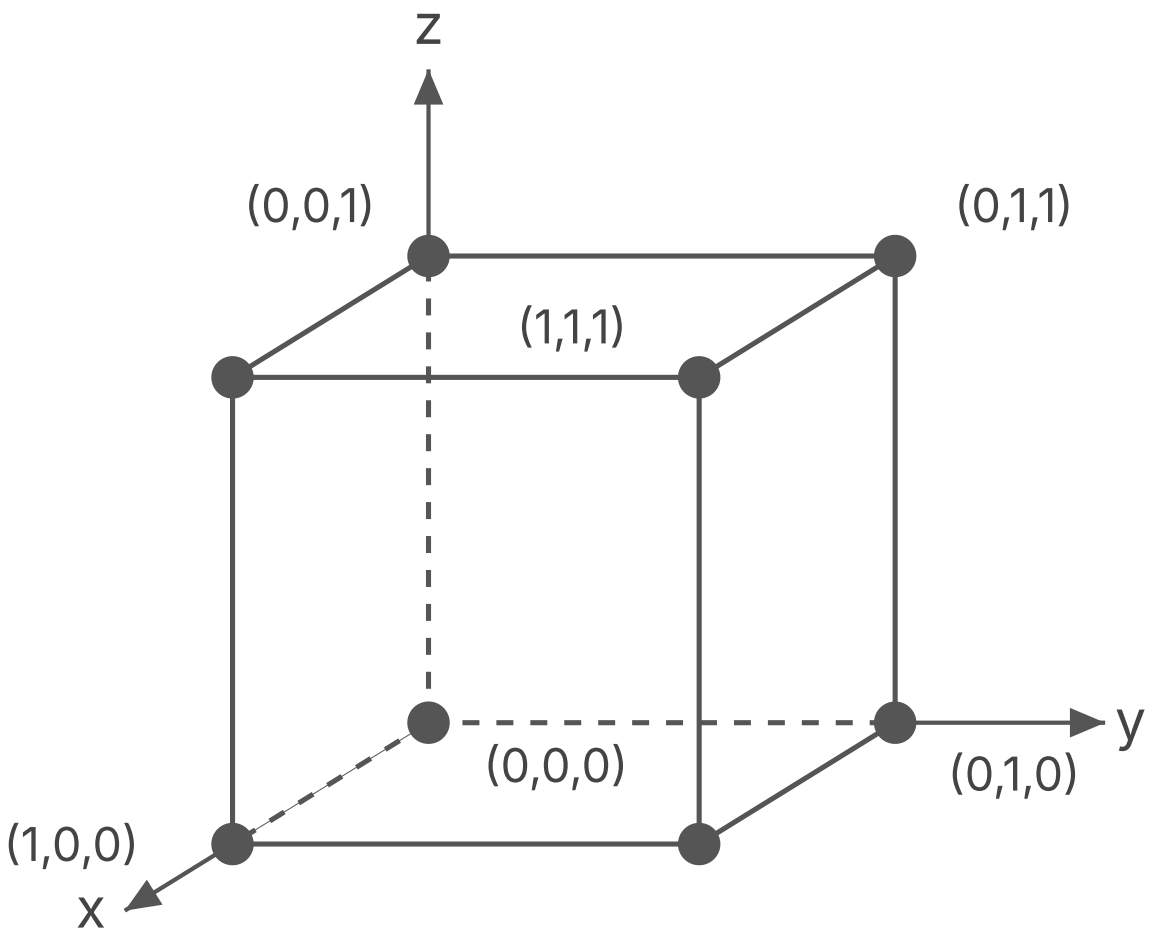


Figure 5. Atomic positions in a cubic cell.

§ 10 Directions in the cubic cell [Session 3 Slides 39-42]

10.1 What a direction is

A **crystallographic direction** is a vector joining two points of the lattice. It is written with **square brackets** and **no commas**, $[u\,v\,w]$. The indices u, v, w are the components of the vector along the three axes, **reduced to the smallest integers** that keep the same ratio.

Two vectors that are **parallel** have the **same indices**: a direction does not depend on where it starts, only on where it points. For that reason any direction can be drawn starting at a corner of the cell.

Negative components are written with a bar over the number: $[\bar{u}\,v\,w]$ means u is negative. ($[\bar{1}\,1\,0]$ is read "minus one, one, zero".)

10.2 How to draw a direction from its indices

1. **Choose the origin.** Normally $O = (0, 0, 0)$. If an index is negative, use a new origin O' (see 10.3).
2. **Scale the indices** so that the largest becomes 1. This keeps the arrow inside the cell. Example: $[0\,1\,2] \to \frac{1}{2}[0\,1\,2] = [0, \frac{1}{2}, 1]$.
3. **Locate the tip:** start at the origin and move those amounts along x, y and z.
4. **Draw the arrow** from the origin to the tip.

Example 1: $[1\,0\,0]$**,** $[1\,1\,0]$**,** $[1\,1\,1]$**.** The largest index is already 1, so the tips are directly $(1, 0, 0)$, $(1, 1, 0)$ and $(1, 1, 1)$: a cube edge, a face diagonal and the body diagonal.

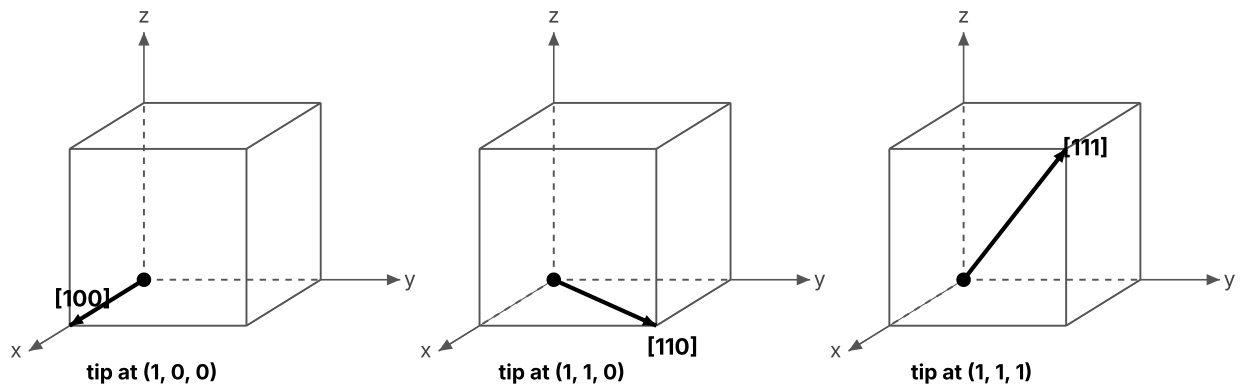


Figure 6. Directions [100], [110] and [111].

Example 2: $[0\,1\,2]$**.**

1. Origin $O = (0, 0, 0)$ (no negative indices).
2. Largest index 2, divide by 2: $[0, \frac{1}{2}, 1]$.
3. Tip at $x = 0, y = \frac{1}{2}, z = 1$: the middle of the top edge on the back face.
4. Arrow from $(0, 0, 0)$ to $(0, \frac{1}{2}, 1)$.

Example 3: $[2\,1\,0]$**.** Divide by 2: $[1, \frac{1}{2}, 0]$. The arrow goes from the origin to the middle of the bottom edge on the front face, $(1, \frac{1}{2}, 0)$. It lies in the bottom face because $w = 0$.

Example 4: $[1\,2\,2]$**.** Divide by 2: $[\frac{1}{2}, 1, 1]$. The tip is at the middle of the top edge on the right face, $(\frac{1}{2}, 1, 1)$.

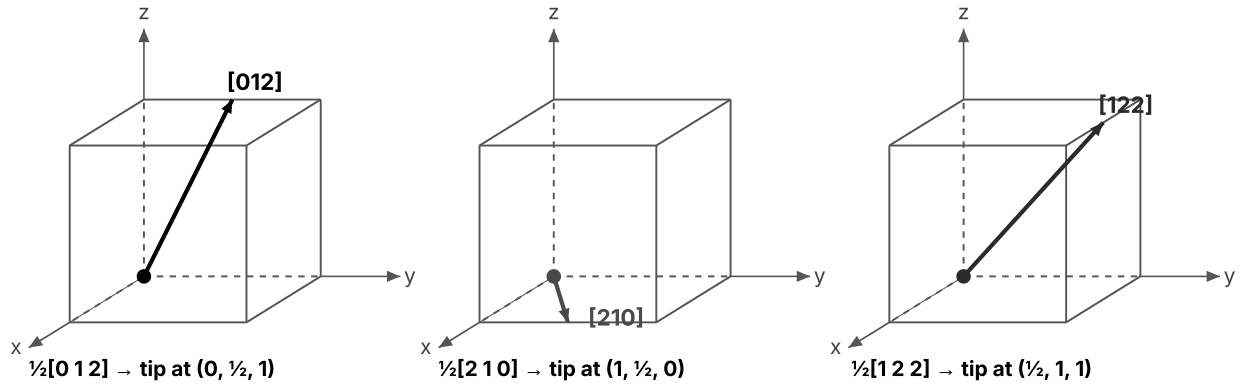


Figure 7. Directions that need reducing: [012], [210], [122].

10.3 Directions with negative indices

If an index is negative, the arrow would leave the cell when drawn from $(0,0,0)$. To keep it inside, move the origin: **the new origin O' has a 1 where the index is negative and a 0 where it is positive (or zero)**. Then draw the arrow from O' as usual.

Direction	New origin O'	Tip (O' + scaled indices)
$[\bar{1}\,1\,1]$	$(1,0,0)$	$(1-1,\ 0+1,\ 0+1) = (0,1,1)$
$[1\,\bar{1}\,0]$	$(0,1,0)$	$(0+1,\ 1-1,\ 0) = (1,0,0)$
$[0\,2\,\bar{1}]$	$(0,0,1)$	scaled $[0,1,-\frac{1}{2}]$: $(0,\ 1,\ 1-\frac{1}{2}) = (0,1,\frac{1}{2})$

Example 5: $[\bar{1}\,2\,2]$**.** The u index is negative, so $O' = (1,0,0)$. Scale by ½: $[-\frac{1}{2}, 1, 1]$. Tip $= (1-\frac{1}{2},\ 0+1,\ 0+1) = (\frac{1}{2}, 1, 1)$. The arrow goes from the front-bottom-left corner $(1,0,0)$ up to the middle of the top edge on the right face.

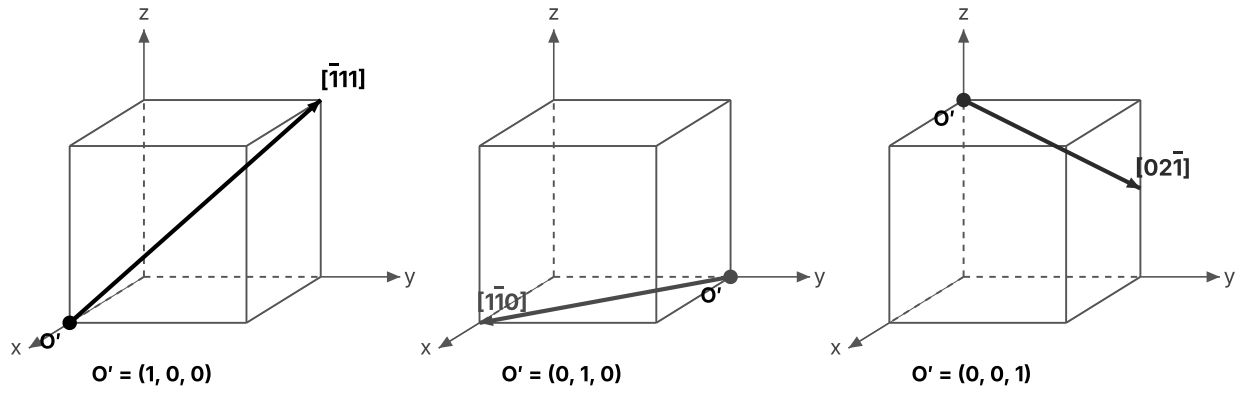


Figure 8. Directions with negative indices and their new origins.

10.4 How to read the indices of a drawn direction

1. Take a vector **along the direction** that passes through the origin (or any parallel copy of it: parallel vectors have the same indices).
2. Measure its **projections** on the three axes, in units of the cell edges a, b, c. In practice: coordinates of the **head minus** coordinates of the **tail**.
3. **Multiply or divide** by a common factor to reach the **smallest integers**.
4. Write them in square brackets without commas, with a bar over any negative number.

Example 6. A vector goes from the tail $(1, 0, 0)$ to the head $(0, 1, 1)$. Head − tail $= (0-1,\ 1-0,\ 1-0) = (-1, 1, 1)$. They are already the smallest integers, so the direction is $[\bar{1}\,1\,1]$.

Example 7. A vector goes from $(0, 1, 1)$ to $(0, 0, 0)$. Head − tail $= (0, -1, -1)$: the direction is $[0\,\bar{1}\,\bar{1}]$. Its opposite, from $(0, 0, 0)$ to $(0, 1, 1)$, is $[0\,1\,1]$: reversing an arrow changes the sign of every index.

Example 8. A vector goes from $(0, 0, 0)$ to $(\frac{1}{2}, 1, 0)$. Head − tail $= (\frac{1}{2}, 1, 0)$. Multiply by 2 to clear the fraction: $[1\,2\,0]$.

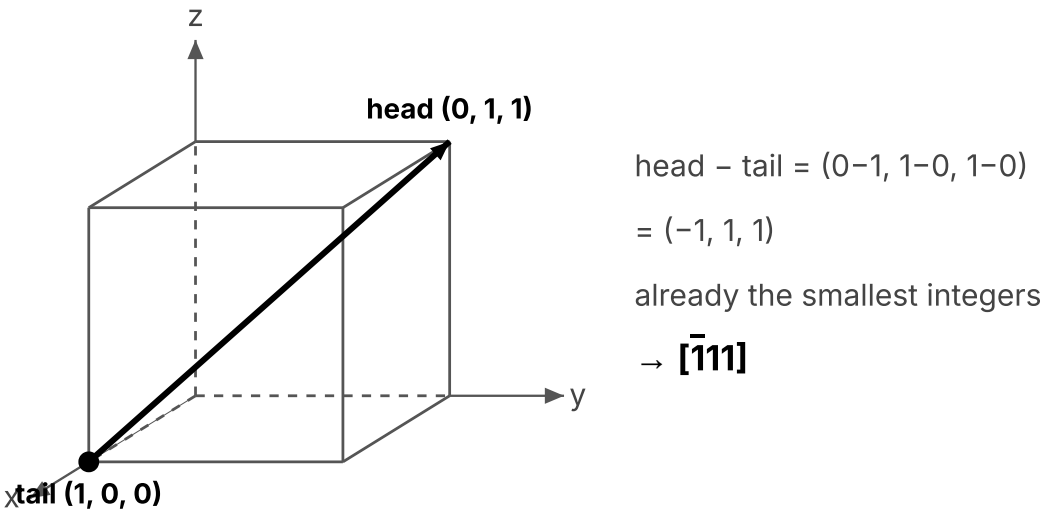


Figure 9. Reading the indices of a drawn direction.

10.5 Families of directions

Directions that are **equivalent** in the lattice (the atoms are spaced the same way along them) form a **family**, written with angle brackets $\langle u\,v\,w \rangle$. In a cube:

- **Cube edges:** $[100]$, $[010]$, $[001]$, $[\bar{1}00]$, $[0\bar{1}0]$, $[00\bar{1}]$ form $\langle 100 \rangle$.
- **Face diagonals:** $\langle 110 \rangle$.
- **Body diagonals:** $\langle 111 \rangle$: the four diagonals $[111]$, $[\bar{1}11]$, $[1\bar{1}1]$, $[11\bar{1}]$ (each with its opposite).

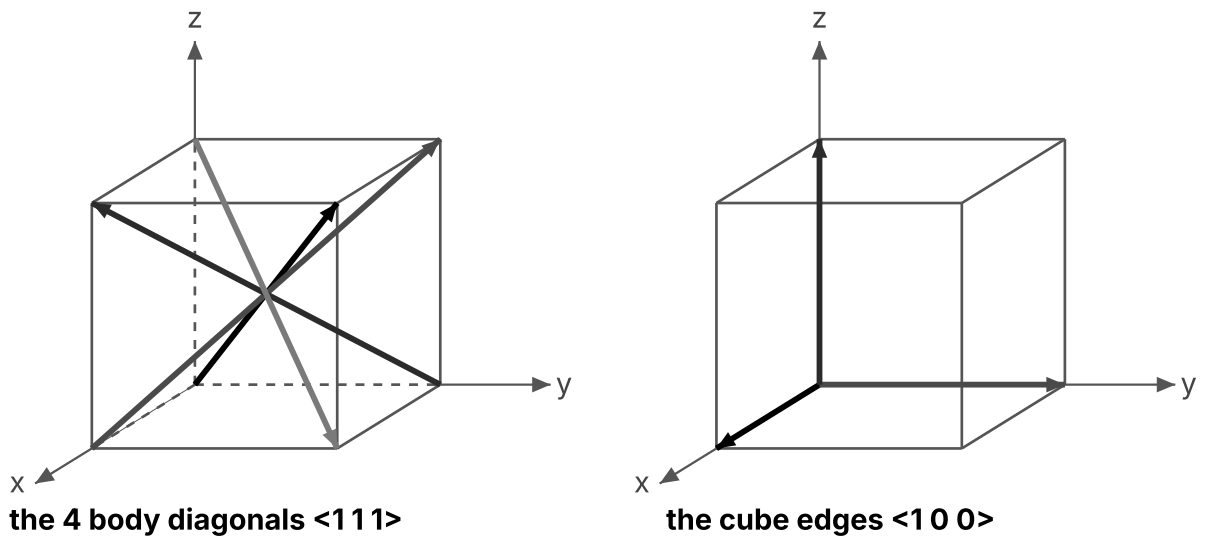


Figure 10. Families of directions <111> and <100>.

§ 11 Planes in the cubic cell [Session 3 Slides 44–50]

11.1 Miller indices of a plane

A crystallographic plane is described by its **Miller indices** $(h\,k\,l)$, written in **round brackets** and **no commas**. They are the **reciprocals of the points where the plane cuts the axes** x, y and z (measured in units of a, b, c):

$$h = \frac{1}{\text{intercept on } x}, \qquad k = \frac{1}{\text{intercept on } y}, \qquad l = \frac{1}{\text{intercept on } z}$$

A plane **parallel** to an axis never cuts it: its intercept is ∞, and $1/\infty = 0$. So a zero index always means "parallel to that axis".

Example: a plane that cuts x at 1 and y at 1 and is parallel to z has intercepts $(1, 1, \infty)$, reciprocals $(1, 1, 0)$: it is the plane $(1\,1\,0)$.

11.2 How to find the indices of a drawn plane

1. **Choose the origin** O. The plane must **not pass through it**; if it does, use the nearest other corner.
2. Find the **intercepts** of the plane with the three axes.
3. Take their **reciprocals**.
4. Reduce them to the **smallest integers** with the same ratio.
5. Write them in **parentheses without commas**: $(h\,k\,l)$.

Intercepts x, y, z	Reciprocals	Plane
1, 1, 1	1, 1, 1	(1 1 1)

Intercepts x, y, z	Reciprocals	Plane
$1,\ 1,\ \infty$	$1,\ 1,\ 0$	$(1\,1\,0)$
$1,\ \infty,\ \infty$	$1,\ 0,\ 0$	$(1\,0\,0)$
$\frac{1}{2},\ \infty,\ \infty$	$2,\ 0,\ 0$	$(2\,0\,0)$
$\frac{1}{2},\ 1,\ \infty$	$2,\ 1,\ 0$	$(2\,1\,0)$
$\infty,\ 1,\ 1$	$0,\ 1,\ 1$	$(0\,1\,1)$

Example 1: $(1\,0\,0)$**,** $(1\,1\,0)$**,** $(1\,1\,1)$**.**

- $(1\,0\,0)$ cuts only x, at 1, and is parallel to y and z: it is the **front face** of the cube.
- $(1\,1\,0)$ cuts x at 1 and y at 1 and is parallel to z: a **vertical diagonal plane** through two opposite vertical edges.
- $(1\,1\,1)$ cuts all three axes at 1: the **triangle** joining $(1, 0, 0)$, $(0, 1, 0)$ and $(0, 0, 1)$.

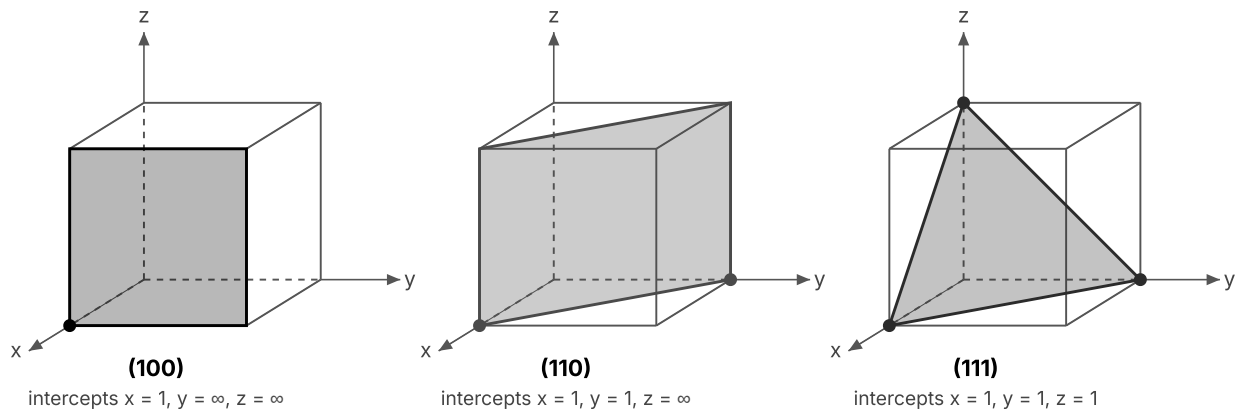


Figure 11. Planes (100), (110) and (111).

Example 2: $(2\,0\,0)$**,** $(2\,1\,0)$**,** $(0\,1\,1)$**.**

- $(2\,0\,0)$: intercept on x is $\frac{1}{2}$, parallel to y and z. It has the same orientation as $(1\,0\,0)$ but cuts the cell **half way**; here the index is kept as 2 to show that position.
- $(2\,1\,0)$: intercepts $x = \frac{1}{2}, y = 1$, parallel to z. A vertical plane that cuts the left face ($y = 0$) along the line $x = \frac{1}{2}$ and reaches the back-right vertical edge ($x = 0, y = 1$).
- $(0\,1\,1)$: parallel to x, cuts y and z at 1. A tilted plane that runs along x, from the top edge of the left face ($y = 0, z = 1$) down to the bottom edge of the right face ($y = 1, z = 0$).

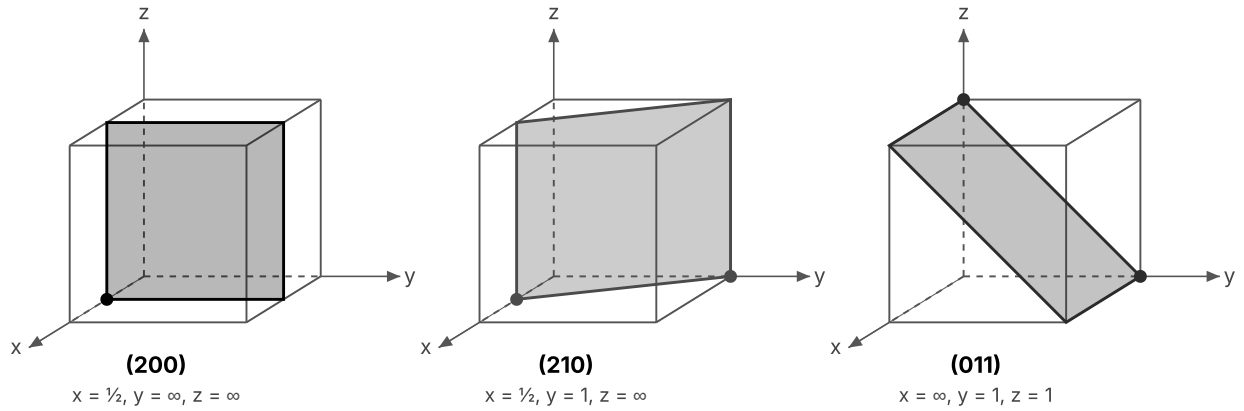


Figure 12. Planes (200), (210) and (011).

11.3 How to draw a plane from its indices

1. **Choose the origin**: $O = (0, 0, 0)$, or O' if any index is negative (see 11.4).
2. Take the **reciprocals** of the indices, $1/h$, $1/k$, $1/l$: these are the **intercepts**.
3. **Mark** the intercepts on the axes ($1/0 = \infty$ means the plane does not cut that axis: it is parallel to it).
4. **Join** the intercepts to draw the plane.

Example 3: $(0\,1\,1)$**.**

1. No negative indices: $O = (0, 0, 0)$.
2. Intercepts: $x = 1/0 = \infty$, $y = 1/1 = 1$, $z = 1/1 = 1$.
3. The plane is parallel to x and passes through $(0, 1, 0)$ and $(0, 0, 1)$.
4. Because it is parallel to x, slide that line along x: the plane is the rectangle with corners $(0, 1, 0)$, $(0, 0, 1)$, $(1, 0, 1)$, $(1, 1, 0)$.

11.4 Planes with negative indices

To draw a plane with a negative index inside the cell, **move the origin** in the direction of that index: as with directions, O' has a **1 in the position of each negative index** and a 0 elsewhere. The intercepts are then measured from O', and a negative intercept means going **backwards** along that axis.

Plane	New origin O'	Intercepts from O'	Result
$(\bar{1}\,0\,0)$	$(1, 0, 0)$	$x = -1$, $y = \infty$, $z = \infty$	the back face $x = 0$
$(1\,\bar{1}\,0)$	$(0, 1, 0)$	$x = 1$, $y = -1$, $z = \infty$	vertical diagonal plane through the edges at $(0, 0)$ and $(1, 1)$
$(1\,\bar{1}\,\bar{1})$	$(0, 1, 1)$	$x = 1$, $y = -1$, $z = -1$	triangle through $(1, 1, 1)$, $(0, 0, 1)$ and $(0, 1, 0)$
$(\bar{2}\,1\,1)$	$(1, 0, 0)$	$x = -\frac{1}{2}$, $y = 1$, $z = 1$	triangle through $(\frac{1}{2}, 0, 0)$, $(1, 1, 0)$ and $(1, 0, 1)$

Example 4: $(1\,\bar{1}\,\bar{1})$**, step by step.**

1. Negative indices in y and z: $O' = (0, 1, 1)$.
2. Intercepts from O': $x = 1/1 = 1$, $y = 1/(-1) = -1$, $z = 1/(-1) = -1$.

3. From $O' = (0, 1, 1)$: one unit forward in x gives $(1, 1, 1)$; one unit back in y gives $(0, 0, 1)$; one unit back in z gives $(0, 1, 0)$.
4. The plane is the triangle joining those three points.

Example 5: $(\bar{2}\,1\,1)$, step by step.

1. Negative index in x: $O' = (1, 0, 0)$.
2. Intercepts from O': $x = 1/(-2) = -\frac{1}{2}, y = 1, z = 1$.
3. From $(1, 0, 0)$: half a unit back in x gives $(\frac{1}{2}, 0, 0)$; one unit in y gives $(1, 1, 0)$; one unit in z gives $(1, 0, 1)$.
4. Join the three points.

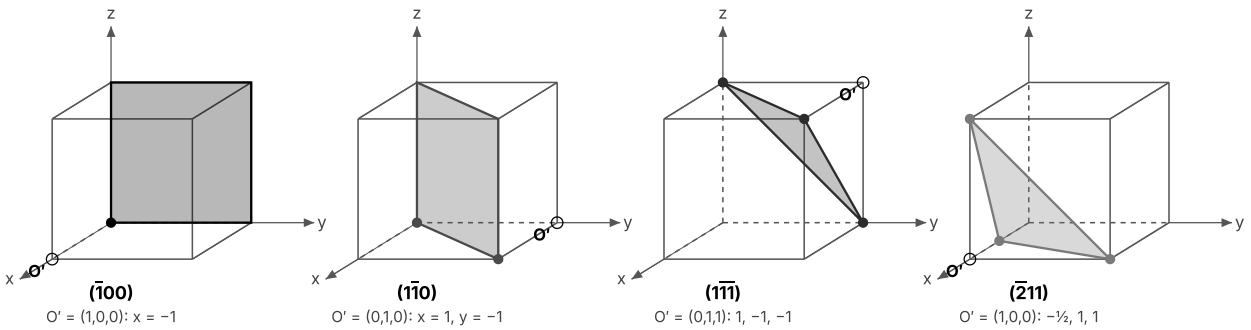


Figure 13. Planes with negative indices and their new origins.

11.5 Families of planes

- **Parallel, equally spaced** planes have the **same Miller indices**.
- Planes that are **equivalent** in the lattice (with the same arrangement of atoms) form a **family**, written with curly brackets $\{h\,k\,l\}$.

$$(100)(010)(001)(\bar{1}00)(0\bar{1}0)(00\bar{1}) = \{100\} \quad \text{(the six cube faces)}$$

$$(111)(\bar{1}11)(1\bar{1}1)(11\bar{1})(\bar{1}\bar{1}1)(\bar{1}1\bar{1})(1\bar{1}\bar{1})(\bar{1}\bar{1}\bar{1}) = \{111\}$$

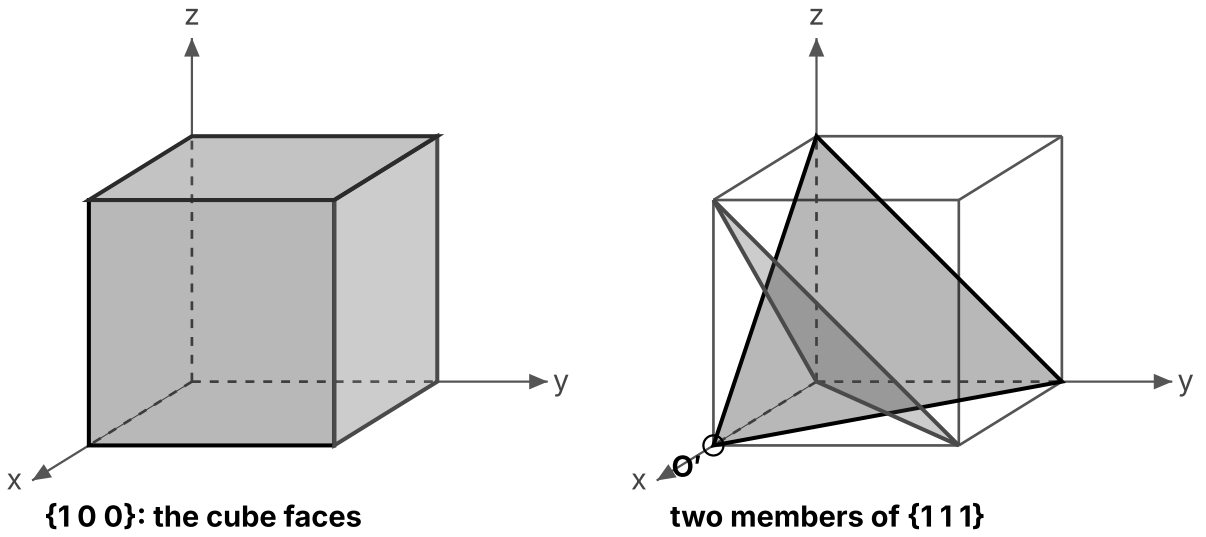


Figure 14. Families of planes {100} and {111}.

11.6 Directions and planes with the same indices

In **cubic** lattices, a direction is **perpendicular** to the plane with the same indices: $[1\,1\,1] \perp (1\,1\,1)$, $[1\,0\,0] \perp (1\,0\,0)$, $[1\,1\,0] \perp (1\,1\,0)$. This only holds for cubic cells.

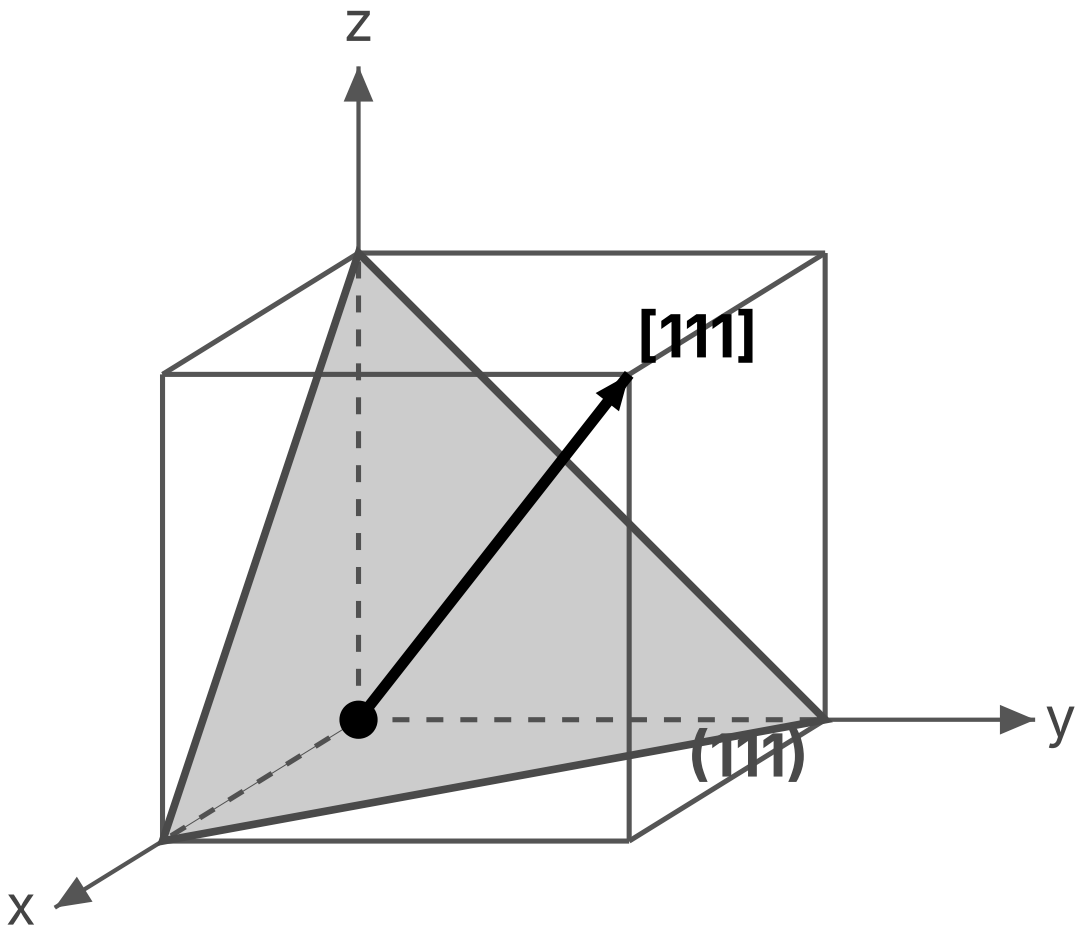


Figure 15. In a cube the [111] direction is perpendicular to the (111) plane.

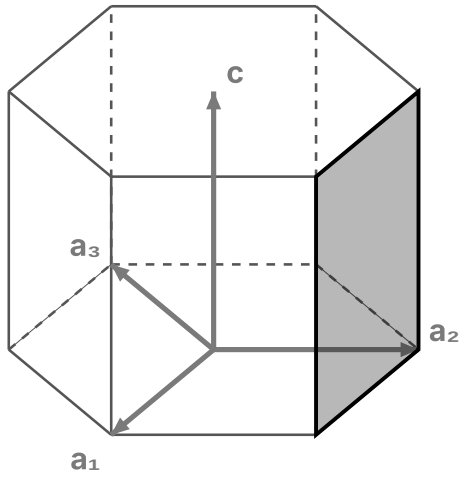
§ 12 Directions and planes in the hexagonal cell [Session 3 Slides 51-52]

The hexagonal cell is described with **four axes**: three in the basal plane, a_1, a_2, a_3, at 120° from each other, and the vertical axis c. Indices therefore have **four numbers**: h, k, i for the three basal axes and l for c.

Planes. Only three of the four indices are independent, because the three basal axes are related:

$$i = -(h + k)$$

So a plane written with three indices $(h'\,k'\,l')$ becomes $(h\,k\,i\,l)$ by adding i. Example: $(0\,1\,0) \equiv (0\,1\,\bar{1}\,0)$, since $i = -(0 + 1) = -1$. This plane is a side face of the prism, parallel to a_1 and to c.



4 axes: a_1, a_2, a_3
(120° apart, in the base)
and c (vertical)

shaded face:
$a_1 = \infty$, $a_2 = 1$, $a_3 = -1$, $c = \infty$
$(01\bar{1}0) \equiv (010)$
check: i = −(h + k)
= −(0 + 1) = −1

Figure 16. Hexagonal cell: the four axes and the $(01\bar{1}0)$ plane.

Directions. A direction $[h'\,k'\,l']$ written in three indices becomes $[h\,k\,i\,l]$ with

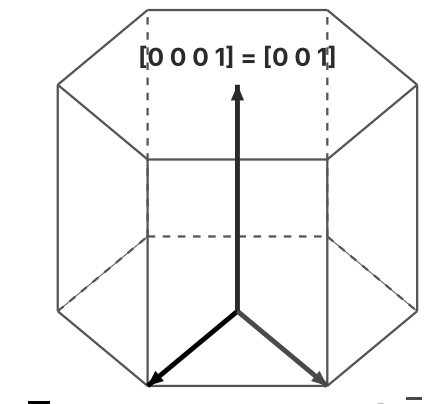
$$h = \tfrac{1}{3}(2h' - k'), \qquad k = \tfrac{1}{3}(2k' - h'), \qquad i = -\tfrac{1}{3}(h' + k'), \qquad l = l'$$

and the result is then reduced to integers.

Example: $[1\,0\,0]$. $h = \frac{1}{3}(2\cdot 1 - 0) = \frac{2}{3}$, $k = \frac{1}{3}(2\cdot 0 - 1) = -\frac{1}{3}$, $i = -\frac{1}{3}(1 + 0) = -\frac{1}{3}$, $l = 0$. Multiplying by 3: $[1\,0\,0] \equiv [2\,\bar{1}\,\bar{1}\,0]$, along a_1.

Example: $[1\,1\,0]$. $h = \frac{1}{3}(2 - 1) = \frac{1}{3}$, $k = \frac{1}{3}(2 - 1) = \frac{1}{3}$, $i = -\frac{1}{3}(1 + 1) = -\frac{2}{3}$, $l = 0$. Multiplying by 3: $[1\,1\,0] \equiv [1\,1\,\bar{2}\,0]$.

Example: $[0\,0\,1]$. $h = k = i = 0$ and $l = 1$: $[0\,0\,0\,1]$, along c.



[1 0 0] → h = ⅓(2·1 − 0) = ⅔
k = ⅓(2·0 − 1) = −⅓, i = −⅓(1 + 0) = −⅓
×3 → [2$\bar{1}\bar{1}$0]

[1 1 0] → h = ⅓, k = ⅓, i = −⅔
×3 → [11$\bar{2}$0]

Figure 17. Directions in the hexagonal cell in 4-index notation.

NOTE ON THE OFFICIAL MATERIAL

The conversion rule for the fourth index is printed as $l = -l'$. With that sign, the vertical direction $[0\,0\,1]$ would become $[0\,0\,0\,\bar{1}]$, pointing down, while the same material labels it $[0\,0\,0\,1]$. The rule is $l = l'$.

§ 13 Spacing between parallel planes [Session 3 Slide 53]

The **interplanar spacing** d_{hkl} is the distance between two neighbouring parallel planes $(h\,k\,l)$. It depends on the Miller indices and on the lattice parameters.

Cubic crystals:

$$d_{hkl} = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$$

Example: for $(1\,1\,0)$, $d_{110} = a/\sqrt{1+1+0} = a/\sqrt{2}$.

Orthogonal crystals in general (all angles 90°, edges a, b, c):

$$\frac{1}{d_{hkl}^2} = \frac{h^2}{a^2} + \frac{k^2}{b^2} + \frac{l^2}{c^2}$$

With $a = b = c$ this reduces to the cubic formula.

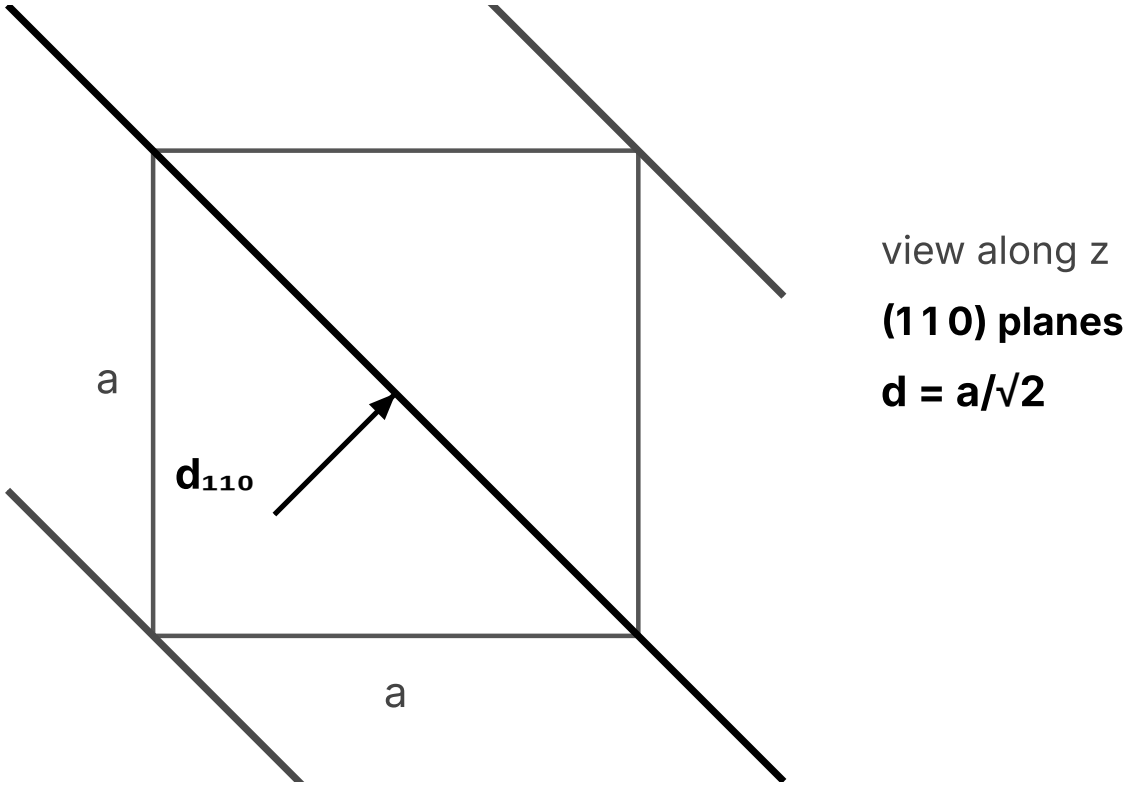


Figure 18. Spacing between parallel (110) planes.

Part II: Densities and crystalline defects

§ 14 Atomic densities [Session 4 Slides 7-9]

14.1 Volume density [Session 4 Slide 7]

$$\rho = \frac{\text{mass of the unit cell}}{\text{volume of the unit cell}} = \frac{n \cdot M}{V_{cell} \cdot N_A}$$

n = atoms per cell, M = atomic weight, V_{cell} = volume of the cell, N_A = Avogadro's number (6.023×10^{23} atoms/mol). Dividing only the number of atoms by the volume gives the number of atoms per unit volume, $\rho_V = n / V_{cell}$.

Example: copper (FCC), $M = 63.5$ g/mol, $R = 1.28$ Å $= 1.28 \times 10^{-8}$ cm.

- Atoms per cell: $n = 8 \cdot \frac{1}{8} + 6 \cdot \frac{1}{2} = 4$.
- Edge: $a = 2\sqrt{2}\,R$, so $V_{cell} = a^3 = (2\sqrt{2}\,R)^3 = 16\sqrt{2}\,R^3$.
- Atoms per volume: $\rho_V = \dfrac{4}{(2\sqrt{2} \cdot 1.28 \times 10^{-8}\ \text{cm})^3} = 8.43 \times 10^{22}\ \text{atoms/cm}^3$.
- Density: $\rho = \dfrac{4 \cdot 63.5\ \text{g/mol}}{6.023 \times 10^{23}\ \text{atoms/mol}} \cdot \dfrac{1}{V_{cell}} = 8.89\ \text{g/cm}^3$.

14.2 Linear density [Session 4 Slide 8]

$$\rho_{[uvw]} = \frac{\text{number of atoms whose centre lies on the line}}{\text{length of the line}} \quad [\text{atoms/m}]$$

Only atoms **cut through their centre** by the line count, and an atom at the end of the segment counts ½ (it is shared with the next segment).

Example: Cu (FCC), direction $[1\,0\,0]$**.** The cube edge has an atom at each end: 2 × ½ = 1 atom, over the length $a = 2\sqrt{2}\,R$:

$$\rho_{[100]} = \frac{1 \text{ atom}}{2\sqrt{2} \cdot 1.28 \times 10^{-8} \text{ cm}} = 2.76 \times 10^{7} \text{ atoms/cm}$$

Example: Cu (FCC), direction $[1\,1\,1]$**.** The body diagonal has length $\sqrt{3}\,a$. In an FCC cell no atom sits at the centre of the cube, so the only atoms on this line are the two corners: 2 × ½ = 1 atom:

$$\rho_{[111]} = \frac{1 \text{ atom}}{\sqrt{3}\,a}$$

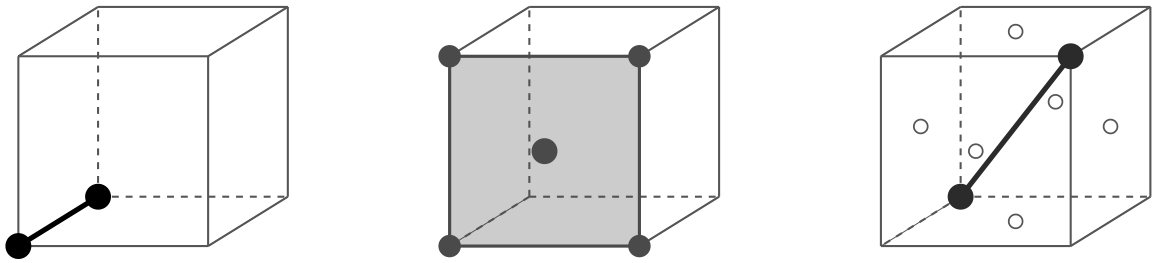
> **NOTE ON THE OFFICIAL MATERIAL**
> The official example counts "1 central atom + 2 · ½ vertices" along $[1\,1\,1]$ in FCC copper, giving $2/(\sqrt{3}\,a) = 2\sqrt{3}/(3a)$. A centre atom only exists in BCC. In FCC the face-centre atoms (such as $(\frac{1}{2}, \frac{1}{2}, 0)$) do not lie on the line (t, t, t), so the count is 1 atom and $\rho_{[111]} = 1/(\sqrt{3}\,a)$.

14.3 Planar density [Session 4 Slide 9]

$$\rho_{(hkl)} = \frac{\text{number of atoms whose centre lies on the plane}}{\text{area of the plane}} \quad [\text{atoms/m}^2]$$

Example: Cu (FCC), plane $(1\,0\,0)$**.** The plane is a cube face: 1 atom at its centre plus 4 corners, each shared by 4 faces (¼ each): 1 + 4 · ¼ = 2 atoms, over the area a^2:

$$\rho_{(100)} = \frac{2 \text{ atoms}}{a^2} = \frac{2}{(2\sqrt{2} \cdot 1.28 \times 10^{-8} \text{ cm})^2} = 1.5 \times 10^{15} \text{ atoms/cm}^2$$



[1 0 0] in FCC
2 × ½ = 1 atom over length a

(1 0 0) face in FCC
1 + 4 × ¼ = 2 atoms over area a²

[1 1 1] in FCC
only 2 × ½ = 1 atom over √3·a

Figure 19. Linear and planar density examples in FCC.

14.4 Summary of the three structures [Session 4 Slide 10]

	FCC	BCC	HCP
Atoms per cell	$8 \cdot \frac{1}{8} + 6 \cdot \frac{1}{2} = 4$	$8 \cdot \frac{1}{8} + 1 = 2$	$12 \cdot \frac{1}{6} + 2 \cdot \frac{1}{2} + 3 = 6$
Atomic radius	$r = \frac{a}{2\sqrt{2}}$	$r = \frac{a\sqrt{3}}{4}$	$r = \frac{1}{2}\sqrt{\frac{a^2}{3} + \frac{c^2}{4}}$
Cell volume	a^3	a^3	$a^2 c \, \frac{3\sqrt{3}}{2}$
Empty volume $V_h = V_c - V_{atoms}$	$5.872\, r^3$	$3.94\, r^3$	$8.808\, r^3$
Empty fraction V_h / V_c	0.26	0.32	0.26
Octahedral sites	4, radius $\frac{1}{2}(a - 2r) = 0.41\, r$	6, radius $\frac{1}{2}(a - 2r) = 0.15\, r$	6, radius $0.41\, r$
Tetrahedral sites	8, radius $\left(\sqrt{\frac{3}{2}} - 1\right) r = 0.225\, r$	12, radius $\left(\sqrt{\frac{5}{3}} - 1\right) r = 0.291\, r$	12, radius $0.225\, r$
Close-packed planes	$\{111\}$: 4	$\{110\}$: 6	$\{0001\}$: 1
Close-packed directions	$\langle 110 \rangle$: 6	$\langle 111 \rangle$: 4	$\langle 2\bar{1}\bar{1}0 \rangle$: 3

The empty fractions match the packing factors: $1 - 0.74 = 0.26$ and $1 - 0.68 = 0.32$.

§ 15 Perfect and real crystals [Session 4 Slides 12-13]

A **perfect crystal** has every atom in its regular position, the same spacing d_{hkl} everywhere for each set of planes, and no missing or extra atoms. It could only exist at 0 K.

A **real crystal** is different: its atoms vibrate, some positions are empty (**vacancies**), some atoms are displaced from their ideal positions, and these **defects** change the properties.

A **crystalline defect** is any disruption of the perfect periodic arrangement of atoms or ions.

Why defects always exist. Creating a defect costs energy (its enthalpy ΔH is positive) but it also increases disorder (its entropy ΔS is positive). The crystal settles where its Gibbs energy is lowest:

$$\Delta G = \Delta H - T\Delta S$$

A defect forms when it lowers G, that is when $\Delta G < 0$, or $\Delta H < T\Delta S$. The equilibrium number of defects is where G stops decreasing ($\Delta G = 0$): beyond it, extra defects would cost more energy than the disorder they add.

The entropy comes from the number of ways W of placing n defects among N possible positions:

$$S = k_B \ln W, \qquad W = \frac{N!}{(N-n)!\,n!}$$

with $k_B = 1.380622 \times 10^{-23}$ J/K, the Boltzmann constant. In one mole of a crystal there are N_A possible places for a single vacancy, so $W \sim 10^{23}$ even for one defect: the gain in entropy is large. Because the $T\Delta S$ term grows with temperature, **the number of defects increases with temperature**.

§ 16 Types of defects [Session 4 Slide 14]

Defects are classified by their **dimension**:

Type	Examples
1. Point (0-D)	vacancies, interstitials, substitutional atoms, Schottky and Frenkel defects, order-disorder
2. Linear (1-D): dislocations	edge (Taylor), screw (Burgers), mixed
3. Planar or extended (2-D)	external surfaces, grain boundaries, twin boundaries, stacking faults
4. Complex (volume, 3-D)	clusters

§ 17 Point defects [Session 4 Slides 15-18]

17.1 What they are [Session 4 Slide 15]

Point defects are local disruptions of an otherwise perfect arrangement. Although they occupy only one or two sites, their effect is felt over much larger distances in the crystal. They appear when atoms move after gaining energy (for example, when the temperature rises). Foreign atoms can also act as point defects:

- **impurities:** elements or compounds that come from the raw materials or the processing;
- **dopants:** elements or compounds added on purpose, in known amounts.

17.2 Vacancies and interstitials [Session 4 Slide 16]

- **Interstitial:** an atom of the crystal sits in an interstitial site instead of a regular one (**self-interstitial** or interstitialcy).
- **Vacancy:** an atom is missing from its regular position.

The number of vacancies at temperature T is

$$n_v = N \exp\left(-\frac{\Delta H_v}{RT}\right), \qquad N = \frac{N_A\,\rho}{M}$$

Symbol	Meaning
n_v	vacancies per cm³
N	lattice points per cm³
ΔH_v	energy to form one mole of vacancies (J/mol)
n_v/N	fraction of vacant sites
N_A, M, ρ	Avogadro's number, atomic (molecular) weight, density

In metals at equilibrium the fraction of vacancies reaches at most about $n_v/N \approx 10^{-4}$.

Vacancies are produced during solidification (impurities, alloys), by bombarding with high-energy particles, during plastic deformation (processing), and simply by heating: higher temperature → stronger thermal vibrations → more vacancies.

17.3 Schottky and Frenkel defects in ionic crystals [Session 4 Slides 17-18]

In an ionic crystal, removing a single ion would leave the crystal electrically charged. Defects therefore come in combinations that keep it neutral:

- **Schottky defect:** a **pair of vacancies of opposite charge**, one cationic and one anionic.

$$n_S = N \exp\left(-\frac{\Delta H_S}{2RT}\right)$$

- **Frenkel defect:** an ion **moves from its regular position to an interstitial site**, leaving a vacancy behind (a vacancy-interstitial pair).

$$n_F = \sqrt{N N_i}\, \exp\left(-\frac{\Delta H_F}{2RT}\right)$$

ΔH_S and ΔH_F are the energies needed to create each defect, N the number of lattice positions, N_i the number of interstitial positions, T the temperature in kelvin. In a given crystal $\Delta H_S \neq \Delta H_F$, and the defect with the **lower** energy is the one that forms.

Example: Schottky defects in NaCl and MgO.

Crystal	ΔH_S (J per defect)	n_S at 300 K (per mol)	n_S at 1000 K (per mol)
NaCl	3.69×10^{-19}	2.64×10^{4}	9.38×10^{17}
MgO	10.57×10^{-19}	2.12×10^{-32}	1.39×10^{7}

Two conclusions: the number of defects **rises enormously with temperature**, and because ΔH_S(MgO) > ΔH_S(NaCl), vacancies are much **harder to create in MgO**.

CHECKING THE NACL VALUE AT 300 K

Here the energy is given per defect, so the gas constant R is replaced by the Boltzmann constant k_B: $\dfrac{\Delta H_S}{2k_BT} = \dfrac{3.69 \times 10^{-19}}{2 \cdot 1.38 \times 10^{-23} \cdot 300} = 44.6$, and $n_S = 6.02 \times 10^{23} \cdot e^{-44.6} \approx 2.6 \times 10^{4}$ per mol.

§ 18 Solid solutions [Session 4 Slides 19, 21-24]

18.1 What a solid solution is [Session 4 Slide 21]

When two materials are mixed as liquids and then solidified, the result is either a **new phase** or a **solid solution**.

A **solid solution** is a crystalline solid with two or more elements dispersed as atoms in a structure that has **a single phase**. There are two kinds:

- **substitutional:** the solute atoms **replace** atoms of the host;
- **interstitial:** the solute atoms sit in the **interstitial sites** of the host. Only small atoms can do this: H, C, N, B.

18.2 Defects created by a solid solution in ionic solids [Session 4 Slide 22]

When MgO and Al_2O_3 form a solid solution, Al^{3+} ions replace Mg^{2+} ions in the MgO crystal. Each Al^{3+} carries one more positive charge than the Mg^{2+} it replaces, so to keep the crystal neutral some cation sites must stay empty. Three Mg^{2+} (total charge +6) are replaced by two Al^{3+} (also +6) and one vacancy:

$$3\,\mathrm{Mg}^{2+} \;\leftrightarrow\; 2\,\mathrm{Al}^{3+} + 1\ \text{vacancy}$$

18.3 Hume-Rothery rules [Session 4 Slide 23]

The conditions for forming a solid solution:

Substitutional solid solutions:

- solute and solvent must have the **same crystal structure**;
- their **atomic radii must be similar**: difference below 30 % for ceramics and below 15 % for metals; above 15 %, solubility is limited (< 1 %);
- they must have **similar electronegativities**;
- they must have the **same valence**.

Interstitial solid solutions:

- the solute atoms must be **smaller** than the solvent atoms;
- they must have **similar electronegativities**.

Meeting all the conditions does **not guarantee** complete solubility; failing one or more of them gives **partial** solubility.

18.4 Order and disorder [Session 4 Slides 19, 24]

In substitutional solid solutions of elements with similar electronegativities, the two kinds of atoms can be arranged in two ways:

- **disorder:** atoms of one kind occupy the positions of the other, and vice versa, at random;
- **order:** above or below a critical temperature, the atoms return to regular positions, each kind on its own sub-lattice.

Example: the Cu-Au alloy is disordered above 390 °C and ordered below 390 °C.

§ 19 Linear defects: dislocations [Session 4 Slides 26-37]

19.1 What a dislocation is [Session 4 Slide 26]

A **dislocation** is a **line imperfection** in an otherwise perfect crystal. Dislocations can move through the crystal under relatively small forces, and their movement can shift whole crystal planes past each other. They explain:

- why the measured strength and stiffness are lower than the theoretical values;
- **plastic deformation** of metals (their workability and ductility).

They form during solidification, during plastic (permanent) deformation, from concentrations of vacancies, and from atomic disorder in solid solutions. There are three types: **edge** (Taylor), **screw** (Burgers) and **mixed**.

19.2 Edge dislocation [Session 4 Slides 29-30]

An **edge dislocation** is equivalent to inserting an **extra half-plane of atoms** into the crystal. It distorts the lattice and stores energy.

It is characterized by the **Burgers vector b**: the magnitude and direction of the lattice distortion caused by the dislocation. To find it, draw a loop around the dislocation taking the same number of atom spacings in each direction; around a perfect crystal the loop would close, but around a dislocation it does not, and **b is the vector needed to close it**.

For an edge dislocation:

- **b is perpendicular** to the dislocation line;
- **b is parallel** to the direction of movement;
- the energy stored grows with the square of its size: $E \propto |\vec{b}|^2$.

19.3 Screw dislocation [Session 4 Slide 31]

A **screw dislocation** bends some lines of atoms locally, as if a **shear stress** had been applied to one part of the crystal. Following the atoms around the dislocation line traces a **spiral ramp** (a screw). The Burgers vector is again the vector that closes a loop around it, and:

- **b is parallel** to the dislocation line;
- **b is perpendicular** to the direction of movement.

19.4 Mixed dislocation [Session 4 Slide 32]

Real dislocations are usually **mixed**: they change gradually from edge to screw character, with a transition region between the two. The **Burgers vector is the same** along the whole dislocation.

19.5 How dislocations move [Session 4 Slides 33-37]

A metal can be deformed plastically only if its dislocations can move; its **strength and ductility** depend on how easily they move.

Slip is the process by which plastic deformation happens through the motion of dislocations:

- the dislocation moves along a **slip direction** on a **slip plane**;
- the slip direction and the slip plane together form a **slip system**;
- for an **edge** dislocation, **b is parallel** to the slip direction;
- for a **screw** dislocation, **b is perpendicular** to the slip direction;

- the slip direction is the direction of **maximum packing** (highest linear density).

Why the densest direction: the energy needed to move a dislocation is $E \propto |\vec{b}|^2$, and the shortest Burgers vector is found along the close-packed direction. In FCC:

- along the close-packed direction $[1\,1\,0]$, atoms touch, so $b = 2R$ and $E \propto 4R^2$;
- along $[1\,0\,0]$, which is not close-packed, the cube edge is b, and the face diagonal of length $4R$ gives $b^2 + b^2 = (4R)^2$, so $b = 2\sqrt{2}\,R$ and $E \propto 8R^2$.

Slip along $[1\,1\,0]$ costs half the energy.

Slip systems of the three structures:

Structure	Slip plane	Slip direction	Number of slip systems
BCC	$\{110\}$	$\langle \bar{1}11 \rangle$	6 × 2 = 12
FCC	$\{111\}$	$\langle \bar{1}10 \rangle$	4 × 3 = 12
HCP	(0001)	$\langle 2\bar{1}\bar{1}0 \rangle$	1 × 3 = 3

Slip happens on the planes of densest packing and along the directions of highest linear density. **More slip systems → more ductility**: HCP, with only 3 slip systems, has far fewer than BCC or FCC.

§ 20 Planar defects [Session 4 Slides 39-46]

Planar defects are boundaries that separate regions of a material that have the **same crystal structure but different orientation**. There are four kinds: grain boundaries, twin boundaries, stacking faults and external surfaces.

20.1 Grain boundaries [Session 4 Slides 40-42]

When a liquid cools, its atoms lose kinetic energy, bonds become stable, and atoms start to gather into crystals. Crystals start (**nucleate**) at many points at the same time and grow until they meet. The solid ends up as a **polycrystal**: many crystals, or **grains**, each with its own orientation. The **grain boundaries** are the surfaces where grains meet.

- A grain boundary is **2 to 5 atomic distances** wide.
- Atoms are packed less densely there than inside the grains, so boundaries are **high-energy** zones where solid-state reactions and atomic diffusion happen more easily.
- Grain boundaries **block dislocation movement**. With smaller grains there are more boundaries, so:

$$\text{grain size} \downarrow \Rightarrow \text{ mechanical strength} \uparrow$$

20.2 Twin boundaries [Session 4 Slides 43-44]

A **twin boundary** separates two parts of a crystal whose planes are slightly misoriented (**crystal twinning**). It is a special grain boundary where one side is the **mirror image** of the other, with a highly symmetrical interface. Twins form during deformation or heat treatments. They hinder slip, making dislocation movement harder, and so they **increase strength**. Austenitic stainless steel and brass (70 Cu / 30 Zn) are typical examples.

20.3 Stacking faults [Session 4 Slide 45]

A **stacking fault** is an interruption of one or two layers in the stacking sequence of close-packed planes. In FCC, the perfect sequence ABC ABC ABC becomes, for example, ABC **AB** ABC: locally, two layers follow the HCP pattern.

20.4 External surfaces [Session 4 Slide 46]

The **external surface** is where the crystal or grain ends. Surface atoms have fewer neighbours (a lower coordination number) than atoms inside, so the surface has a **higher energy** than the interior.

§ 21 Polymorphism and allotropy [Session 4 Slides 49-50]

- **Polymorphism:** the ability of a solid material to exist in **more than one crystal structure**.
- **Allotropy:** the ability of a chemical **element** to exist in two or more forms in the same physical state.

The same element or compound can take different crystal structures under different conditions of pressure and temperature.

Carbon:

- **diamond:** a three-dimensional covalent network → hard, transparent, electrical insulator;
- **graphite:** a layered structure with secondary bonds between layers → soft, opaque, electrical conductor.

Zirconia (ZrO_2): it changes from monoclinic to tetragonal and then to cubic (CaF_2-type, fcc) as it is heated. The shape change of the transformation consumes energy, and this is used to make the material **tougher** (tetragonal zirconia polycrystal).

Iron (steels): BCC at room temperature, **FCC from 910 °C**. This change of structure is what makes it possible to **heat treat** steels.